Chapter 6: Power Series

6.1 Power Series

Reference: Stewart's Calculus 8E, Section 11.8

Goal: Study how a function can be expressed as an "infinite polynomial' called a power series. The key concept is that of *radius of convergence.*

Definition 1. (Power Series)

A **power series** in x is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots$$

where x is a variable, and the c_n's are constants called the **coefficients** of the series.

NOTE: A power series resembles a polynomial in x. The only difference is that f has infinitely many terms, while a polynomial has only finitely many terms.

Example 6.1. The geometric series

$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$ $(|r|<1)$

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots \Rightarrow \begin{matrix} a=1 \\ r=x \end{matrix}$$

is an example of a power series which converges when $-1 < x < 1$ and diverges when $|x| \geq 1$.

Definition 2. (Power series at $x = a$)

More generally, a series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots$$

is called a **power series in** $(x - a)$ or a **power series at** $x = a$.

centre

* Any power series MUST converge at the centre, namely, it converges to c_0.

power series. since it has the form $\sum_{n=0}^{\infty} c_n(x-a)^n$

Example 6.2. For what values of x is the series $\sum_{n=0}^{\infty} n!x^n$ convergent?

centre $a=0$. $c_n = n!$

Solution. Clearly, the series converges when $x = 0$. Suppose $x \neq 0$. We use the Ratio Test: Let $a_n = n!x^n$. Then

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(n+1)!x^{n+1}}{n!x^n}\right| = \lim_{n\to\infty}(n+1)|x| = \infty.$$

By the Ratio Test, the series diverges whenever $x \neq 0$. So the series converges only when $x = 0$. □

centre

Example 6.3. For what values of x does the series $\sum_{n=1}^{\infty}\frac{(x-3)^n}{n}$ converge?

Idea: Ratio Test (sometimes can use root test)

Solution. Let $a_n = \frac{(x-3)^n}{n}$. Then

$$\begin{aligned}\left|\frac{a_{n+1}}{a_n}\right| &= \left|\frac{(x-3)^{n+1}}{n+1}\cdot\frac{n}{(x-3)^n}\right| \\ &= \frac{|x-3|}{1+\frac{1}{n}} \\ &\longrightarrow |x-3| \quad \text{as } n\to\infty.\end{aligned}$$

By Ratio Test

a) $\rho = |x-3| < 1 \Rightarrow$ absolute converge.

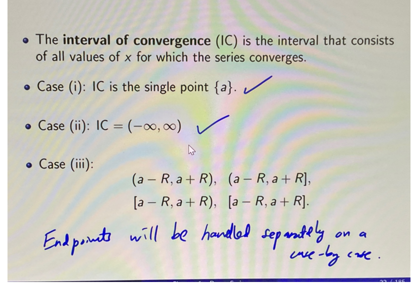
b) $\rho = |x-3| > 1 \Rightarrow$ divergence

By the Ratio Test, the given series is

- absolutely convergent, and is therefore convergent when $|x-3| < 1$, i.e. when $2 < x < 4$
- divergent when $|x-3| > 1$, i.e. when $x < 2$ or $x > 4$.

The Ratio Test has no information when $|x-3| = 1$, i.e. when $x = 2, 4$, so we have to consider these cases separately.

(i) If we substitute $x = 4$ into the series, it becomes $\sum \frac{1}{n}$ the harmonic series, which is known to be divergent.



(ii) If we substitute $x = 2$ into the series, it becomes $\sum \frac{(-1)^n}{n}$ which converges by the Alternating Series Test.

In summary, the interval of convergence of the given power series is $[2, 4)$. □

The last example is an instance of a more general phenomenon about convergence of power series.

> **Theorem 1.** For any given power series $\sum_{n=0}^{\infty} c_n(x-a)^n$, there are only **three** possibilities:
>
> (i) The series converge only when $x = a$.
>
> (ii) The series converges for all x.
>
> extreme cases
>
> (iii) There is a positive number R such that the series converges if $|x - a| < R$, and diverges if $|x - a| > R$.

typical scenario

* End points need to be analysed separately ⇒ like previous example

NOTE:

- The number R is case (iii) is called the **radius of convergence** of the power series.
- By convention, $R = 0$ in case (i), and $R = \infty$ in case (ii).
- The **interval of convergence** (depends on end points) is the interval that consists of all values of x for which the series converges. In case (i), the interval of convergence is the single point a; In case (ii), the interval of convergence is $(-\infty, \infty)$. In case (iii), the inequality $|x - a| < R$ can be rewritten as the interval $(a - R, a + R)$. In the theorem, there is no mention about the convergence at $x = a \pm R$. That is, at $x = a \pm R$, anything can happen, and thus we have four possibilities for the interval of convergence in case (iii):

$$(a - R, a + R), \quad (a - R, a + R], \quad [a - R, a + R), \quad [a - R, a + R].$$

⇒ all have the same radius of convergence
↳ only difference is if endpoints are included.

Example 6.4. Find the radius of convergence and the interval of convergence of the series

centre, $a = 0$

$$\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}}.$$

* everytime you see a power series, identify the centre first.

If for some reason, you obtain
$\rho = |3x-4| < 1$
$3|x-\frac{4}{3}| < 1$
$|x-\frac{4}{3}| < \frac{1}{3}$
$|x-a| < R$
must be of the form $|x-a|$

Solution. Let $a_n = \dfrac{(-3)^n x^n}{\sqrt{n+1}}$. Then

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-3)^{n+1}x^{n+1}}{\sqrt{n+2}} \cdot \frac{\sqrt{n+1}}{(-3)^n x^n}\right|$$
$$= \left|(-3x)\sqrt{\frac{n+1}{n+2}}\right|$$
$$= 3\sqrt{\frac{1+1/n}{1+2/n}}|x|$$
$$\longrightarrow 3|x| \quad \text{as } n \to \infty.$$

Root Test

$a_n = \dfrac{(-3)^n x^n}{\sqrt{n+1}}$

$L = \lim_{n\to\infty} |a_n|^{\frac{1}{n}}$
$= \lim_{n\to\infty} \dfrac{(|-3|^n |x|^n)^{\frac{1}{n}}}{[(n+1)^{\frac{1}{2}}]^{\frac{1}{n}}}$
$= 3|x| \lim_{n\to\infty} \left[(n+1)^{\frac{1}{2n}}\right]$
$= 3|x| \lim_{n\to\infty} e^{\ln(n+1)^{\frac{1}{2n}}}$
$= 3|x| \lim_{n\to\infty} e^{\frac{\ln(n+1)}{2n}}$
$= 3|x| e^{\lim_{x\to\infty} \frac{\ln(x+1)}{2x}}$ → need to do L'Hopital, so change to x to show change to function
$= 3|x| e^{\lim_{n\to\infty} \frac{\left(\frac{1}{x+1}\right)}{2}}$
$= 3|x| e^0$
$= 3|x|$

By the Ratio Test, the series

$|x| < \frac{1}{3} \Rightarrow -\frac{1}{3} < x < \frac{1}{3}$

- converges if $3|x| < 1$, i.e. if $-\frac{1}{3} < x < \frac{1}{3}$.
- diverges if $3|x| > 1$, i.e. if $x > \frac{1}{3}$ or $x < -\frac{1}{3}$.

This means that the radius of convergence is $R = \frac{1}{3}$.

The Ratio Test has no information when $3|x| = 1$, i.e. when $x = \pm\frac{1}{3}$. We consider these cases separately.

(i) If $x = -\frac{1}{3}$, then the series becomes

$$\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}} = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n+1}}$$

which diverges. This is because $\frac{1}{\sqrt{n+1}} \geq \frac{1}{\sqrt{2n}} = \frac{1}{\sqrt{2}}\frac{1}{\sqrt{n}}$, and since $\frac{1}{\sqrt{n}}$ diverges (this is a p-series with $p = 1/2 < 1$), by the Comparison Test, the series $\sum \frac{1}{\sqrt{n+1}}$ diverges.

(ii) If $x = \frac{1}{3}$, then the series becomes

$$\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$$

which converges by the Alternating Series Test.

Is $2x^2+x+1$ a power series?

Yes. $1+x+2x^2 = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \ldots$

$c_0 = 1$
$c_1 = 1$
$c_2 = 2$
$c_3 = c_4 = \ldots = 0$

In summary, the interval of convergence of the given power series is $(-\frac{1}{3}, \frac{1}{3}]$. □

Example 6.5. Find the radius of convergence and the interval of convergence of the series

$$\sum_{n=0}^{\infty} \frac{n(x+2)^n}{3^{n+1}}.$$

Root Test

$$L = \lim_{n\to\infty} |a_n|^{\frac{1}{n}} = \lim_{n\to\infty} \left(\frac{n(x+2)^n}{3^{n+1}}\right)^{\frac{1}{n}} = |x+2| \lim_{n\to\infty} \frac{n^{\frac{1}{n}}}{3 \cdot 3^{\frac{1}{n}}} = \frac{|x+2|}{3}\left(\frac{1}{3}\right) = \frac{|x+2|}{3}$$

Solution. Let $a_n = \dfrac{n(x+2)^n}{3^{n+1}}$. Then

$$\begin{aligned}\left|\frac{a_{n+1}}{a_n}\right| &= \left|\frac{(n+1)(x+2)^{n+1}}{3^{n+2}} \cdot \frac{3^{n+1}}{n(x+2)^n}\right| \\ &= \left(1+\frac{1}{n}\right)\frac{|x+2|}{3} \\ &\longrightarrow \frac{|x+2|}{3} \quad \text{as } n \to \infty.\end{aligned}$$

By the Ratio Test, the series

- converges if $\frac{|x+2|}{3} < 1$, i.e. if $-5 < x < 1$.
- diverges if $\frac{|x+2|}{3} > 1$, i.e. if $x > 1$ or $x < -5$.

centre of power series = −2

This means that the radius of convergence is $R = 3$.

The Ratio Test has no information when $\frac{|x+2|}{3} = 1$, i.e. when $x = -5, 1$. We consider these cases separately.

(i) If $x = -5$, then the series becomes

$$\sum_{n=0}^{\infty} \frac{n(-3)^n}{3^{n+1}} = \frac{1}{3}\sum_{n=0}^{\infty} (-1)^n n$$

which diverges by the nth Term Test for Divergence, since $(-1)^n n$ does not converge to 0.

(ii) If $x = 1$, then the series becomes

$$\sum_{n=0}^{\infty} \frac{n3^n}{3^{n+1}} = \frac{1}{3}\sum_{n=0}^{\infty} n$$

which also diverges by the nth Term Test for Divergence, since n does not converge to 0.

In summary, the interval of convergence of the given power series is $(-5, 1)$. $\square$

6.2 Representation of Function as Power Series

Reference: Stewart's Calculus 8E, Section 11.9

Goal: We learn how to represent certain types of functions as sums of power series by manipulating geometric series or by differentiating or integrating such a series.

Let us start with the geometric series

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n, \tag{6.1}$$

which we know is convergent for all $|x| < 1$.

The idea is to use this series to express a given function as a sum of a power series.

Example 6.6. Express the function $f(x) = \dfrac{1}{1+x^2}$ as the sum of a power series in x and find the interval of convergence.

<u>*Solution.*</u> Replace x by $-x^2$ in Equation (6.1), we have

$$\begin{aligned}
\frac{1}{1+x^2} &= \frac{1}{1-(-x^2)} \\
&= \sum_{n=0}^{\infty} (-x^2)^n \\
&= 1 - x^2 + x^4 - x^6 + x^8 - \cdots
\end{aligned}$$

Since this is a geometric series, it converges when

$$|-x^2| < 1 \iff |x^2| < 1 \iff |x| < 1.$$

Therefore, the interval of convergence is $(-1, 1)$. □

NOTE: We could have determined the radius of convergence by applying the Ratio Test, but that much work is unnecessary here.

Example 6.7. Find a power series representation of $\dfrac{1}{3x+1}$ at $x = 1$, and determine its interval of convergence.

centre

Solution. Again, we shall make use of the geometric series (6.1).

$$
\begin{aligned}
\frac{1}{3x+1} &= \frac{1}{3(x-1)+4} \\
&= \frac{1}{4\left(1-\left(-\frac{3(x-1)}{4}\right)\right)} \\
&= \frac{1}{4}\sum_{n=0}^{\infty}\left(-\frac{3(x-1)}{4}\right)^n \\
&= \sum_{n=0}^{\infty}\frac{(-1)^n 3^n}{4^{n+1}}(x-1)^n.
\end{aligned}
$$

accepted as final answer

centred at $x=1$

The series converges when

$$\left|-\frac{3(x-1)}{4}\right| < 1 \iff |x-1| < \frac{4}{3}.$$

So the interval of convergence is $\left(-\frac{1}{3}, \frac{7}{3}\right)$. $\square$

Example 6.8. Find a power series representation of $\dfrac{x^3}{x+2}$ in x.

means centre is 0

if in $x-1 \Rightarrow$ centre $=1$
if in $x+2 \Rightarrow$ centre $=-2$

Solution. We first find the power series representation of $\dfrac{1}{x+2}$ using the geometric series $\frac{1}{1-x} = 1 + x + x^2 + \cdots$.

$$
\begin{aligned}
\frac{1}{x+2} &= \frac{1}{2}\cdot\frac{1}{1-(-x/2)} \\
&= \frac{1}{2}\sum_{n=0}^{\infty}\left(-\frac{x}{2}\right)^n \\
&= \sum_{n=0}^{\infty}\frac{(-1)^n}{2^{n+1}}x^n.
\end{aligned}
$$

Multiplying by x^3:

$$\begin{aligned}
\frac{x^3}{x+2} &= x^3 \cdot \frac{1}{x+2} \\
&= x^3 \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} x^n \\
&= \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} x^{n+3} \\
&= \sum_{n=3}^{\infty} \frac{(-1)^{n-1}}{2^{n-2}} x^n \quad \text{(after reindexing)} \\
&= (-4) \sum_{n=3}^{\infty} \left(-\frac{x}{2}\right)^n
\end{aligned}$$

The geometric series $\sum_{n=3}^{\infty} \left(-\frac{x}{2}\right)^n$ converges if and only if

$$\left|-\frac{x}{2}\right| < 1 \quad \Longleftrightarrow \quad |x| < 2.$$

So the interval of convergence is $(-2, 2)$. $\square$

Our next theorem states, in essence, that power series are well-behaved functions in the following sense: A power series $f(x)$ is differentiable within its interval of convergence and we may differentiate and integrate $f(x)$ as if it were a polynomial.

Theorem 2. (Term-by-Term Differentiation and Integration of Power Series)

Suppose that the power series

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$$

has radius of convergence $R > 0$. Then $f(x)$ is differentiable on the interval $(a - R, a + R)$, and its derivative and antiderivative may be computed term by term. More precisely,

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + \ldots$$

$$\frac{d}{dx} f(x) = \frac{d}{dx} \sum_{n=0}^{\infty} c_n(x-a)^n$$

can bring in cause "term-by-term"

$$f'(x) = \sum_{n=0}^{\infty} \frac{d}{dx} c_n(x-a)^n = \sum_{n=1}^{\infty} \frac{d}{dx} c_n(x-a)^n = \sum_{n=1}^{\infty} c_n \cdot n \cdot (x-a)^{n-1}$$

when expanded, c_0 constant $\Rightarrow \frac{d}{dx}$ is zero

(i) $$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}$$

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$$

$$\int f(x)\,dx = \int \sum_{n=0}^{\infty} c_n(x-a)^n\,dx = \sum_{n=0}^{\infty} \int c_n(x-a)^n\,dx = \sum_{n=0}^{\infty} \frac{c_n(x-a)^{n+1}}{n+1} + C$$

(ii) $$\int f(x)\,dx = C + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$$

Moreover, the series in (i) and (ii) have the same radius of convergence R.

but may not have same interval of convergence.

$\boxed{\textbf{Example 6.9.}}$ Prove that

$$\frac{1}{(1-x)^2} = \underbrace{1 + 2x + 3x^2 + 4x^3 + 5x^4 + \cdots}_{\sum_{n=0}^{\infty} (n+1)x^n \text{ or } \sum_{n=1}^{\infty} n x^{n-1}}$$

for $-1 < x < 1$.

$$\frac{1}{(1-x)^2} = \frac{d}{dx}\left(\frac{1}{1-x}\right)$$

$$= \frac{d}{dx}\left(\sum_{n=0}^{\infty} x^n\right) \Longrightarrow \text{only if } |x|<1$$

$$= \sum_{n=0}^{\infty} \frac{d}{dx}(x^n)$$

$$= \sum_{n=1}^{\infty} n x^{n-1} \quad \text{if } |x| < 1 \nearrow R$$

Solution. Noting that

$$\frac{d}{dx}\frac{1}{1-x} = \frac{1}{(1-x)^2}$$

we obtain the result by differentiating the geometric series term by term for $|x| < 1$:

$$\begin{aligned} \frac{d}{dx}\frac{1}{1-x} &= \frac{d}{dx}\left(1 + x + x^2 + x^3 + x^4 + \cdots\right) \\ \frac{1}{(1-x)^2} &= 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \cdots \end{aligned}$$

The above expansion is valid for $|x| < 1$ because the geometric series has radius of convergence $R = 1$.

Example 6.10. Prove that for $-1 < x < 1$,

$$\tan^{-1} x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$$

Solution. First, substitute $-x^2$ into x in the geometric series (6.1) to obtain

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \cdots$$

$$\frac{1}{1-(-x^2)}$$

Since the geometric series has radius of convergence $R = 1$, this expansion is valid for $|x^2| < 1$, that is $|x| < 1$.

Recall that $\tan^{-1} x = \displaystyle\int \frac{1}{1+x^2}\,dx$. Now, integrate this series term by term, we have

$$\begin{aligned}
\tan^{-1} x &= \int \frac{1}{1+x^2}\,dx \\
&= \int (1 - x^2 + x^4 - x^6 + \cdots)\,dx \\
&= C + x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots
\end{aligned}$$

where C is a constant.

To determine the constant C, set $x = 0$. We obtain $\tan^{-1} 0 = 0 = C$. This proves the result. □

$$\begin{aligned}
\tan^{-1} x &= \int \frac{1}{1+x^2}\,dx \\
&= \int \frac{1}{1-(-x^2)}\,dx \\
&= \int \sum_{n=0}^{\infty} (-x^2)^n\,dx \\
&= \sum_{n=0}^{\infty} \int (-x^2)^n\,dx \\
&= \sum_{n=0}^{\infty} \int (-1)^n x^{2n}\,dx \\
&= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} + C
\end{aligned}$$

If $x=0$, then,

$0 = 0 + C \Rightarrow C = 0$

Extra Example:

Find the sum of the series $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n} = \sum_{n=1}^{\infty} n^2 \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} n^2 x^n$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \sum_{n=1}^{\infty} n x^{n-1}$$

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1}$$

Multiply both sides by x

$$\frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} n x^n$$

no need change, there's no constant term!

$$\frac{d}{dx} \frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} n^2 x^{n-1}$$

$$\frac{(1-x)^2 \cdot 1 - x \cdot 2(1-x)(-1)}{(1-x)^4} = \sum_{n=1}^{\infty} n^2 x^{n-1}$$

$$\frac{(1-x)[1-x+2x]}{(1-x)^4} = \sum_{n=1}^{\infty} n^2 x^{n-1}$$

$$\frac{x+1}{(1-x)^3} = \sum_{n=1}^{\infty} n^2 x^{n-1}$$

Multiply both sides by x:

$$\frac{x(x+1)}{(1-x)^3} = \sum_{n=1}^{\infty} n^2 x^n$$

Finally sub $x = \frac{1}{2}$

$$6 = \frac{\frac{1}{2}\left(\frac{1}{2}+1\right)}{\left(\frac{1}{2}\right)^3} = \sum_{n=1}^{\infty} n^2 \left(\frac{1}{2}\right)^n$$

6.3 Taylor and Maclaurin Series

Reference: Stewart's Calculus 8E, Section 11.10

> **Goal:** We develop a general method for finding power series representations of functions. The resulting series is called a Taylor series. We study the error in approximating a function by using a Taylor polynomial. If we can show that this error tends to 0 in the limit, then the function will be represented by its Taylor series.

We start by supposing that $f(x)$ has a power series expansion centred at $x = a$ that is valid for all x in an interval $(a - R, a + R)$ with $R > 0$:

$$f(x) \quad = \quad c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + c_4(x-a)^4 + \cdots, \quad |x-a| < R. \tag{6.2}$$

Our goal is to determine the coefficients c_0, c_1, c_2, ….

- c_0:

  We can find c_0 by substituting $x = a$ into the power series (6.2), and so

  $$c_0 = f(a).$$

- c_1:

  By differentiating Equation (6.2) term by term (Theorem 2):

  $$f'(x) \quad = \quad c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + 4c_4(x-a)^3 + \cdots, \quad |x-a| < R. \tag{6.3}$$

  Substituting $x = a$ into Equation (6.3) gives

  $$c_1 = f'(a).$$

- c_2:

  By differentiating Equation (6.3) term by term (Theorem 2):

  $$f^{(2)}(x) \quad = \quad 2c_2 + 3 \cdot 2c_3(x-a) + 4 \cdot 3c_4(x-a)^2 + \cdots, \quad |x-a| < R. \tag{6.4}$$

Substituting $x = a$ into Equation (6.4) gives

$$c_2 = \frac{1}{2} f^{(2)}(a) = \frac{1}{2!} f^{(2)}(a).$$

- c_3:

  By differentiating Equation (6.4) term by term (Theorem 2):

$$f^{(3)}(x) = 3 \cdot 2c_3 + 4 \cdot 3 \cdot 2c_4(x-a) + 5 \cdot 4 \cdot 3c_5(x-a)^2 \cdots, \quad |x-a| < R. \tag{6.5}$$

  Substituting $x = a$ into Equation (6.5) gives

$$c_3 = \frac{1}{3 \cdot 2} f^{(3)}(a) = \frac{1}{3!} f^{(3)}(a)$$

By now you can see the pattern. Solving the equation for the nth coefficient, we get

$$\boxed{c_n = \frac{f^{(n)}(a)}{n!}.}$$

By convention, $0! = 1$ and $f^{(0)} = f$, $f^{(1)} = f'$, and so the formula is valid for all $n = 0, 1, 2, 3, \ldots$.

Theorem 3. (Uniqueness of the Power Series Expansion)

If f has a power series expansion (representation) at $x = a$, that is, if

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n, \quad |x-a| < R, \quad \text{with} \quad R > 0 \tag{6.6}$$

then its coefficients are given by the formula

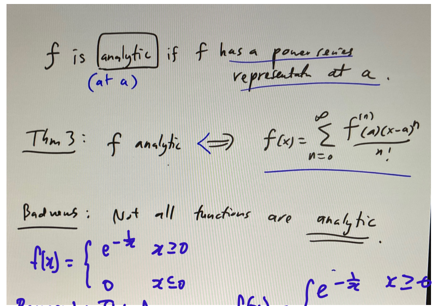
$$c_n = \frac{f^{(n)}(a)}{n!}.$$

Proof is beyond syllabus

- The power series on the right-hand side of Equation (6.6) is called the **Taylor series of $f(x)$ centred at $x = a$.**
- In the special case $a = 0$, the Taylor series is also called the **Maclaurin series**.

If f(x) has a power series representation at a $\Rightarrow f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)} a}{n!} (x-a)^n$

special case of Taylor series.



Remark: This function $f(x) = \begin{cases} e^{-\frac{1}{x}} & x \geq 0 \\ 0 & x \leq 0 \end{cases}$ is not equal to its Taylor series at 0.

IMPORTANT: Theorem (3) starts with the assumption that f has a power series representation valid for $|x-a| < R$. With this assumption, if we want to find this power series representation, then it tells us that the only candidate for the job is the Taylor series. However, without this assumption, one can still compute a Taylor series given f but there is no guarantee that the Taylor series even converge, let alone be equal to the function f!

misconception: Taylor series we obtain is always equal to the function

This is actually not true

Example 6.11.

Find the Maclaurin series of the function $f(x) = e^x$ and its radius of convergence.

Solution. Note that $f^{(n)}(x) = e^x$ for all n. So

$$f^{(n)}(0) = 1, \quad \text{for all } \; n \geq 0.$$

Therefore, the Maclaurin series of f is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots.$$

To find the radius of convergence we let $a_n = \frac{x^n}{n!}$. Then

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \frac{|x|}{n+1} \longrightarrow 0 < 1.$$

So, by Ratio Test, the series converges for all x, and the radius of convergence is $R = \infty$. $\square$

converges for all x

WARNING: Example 6.11 only tells us that the Maclaurin series of e^x converges for all x. It DOES NOT tell us that e^x is equal to its Maclaurin series. By Theorem 3, if e^x has a power series expansion at 0, then it is equal to its Maclaurin series. The question now is

How do we know if e^x does have a power series representation?

To answer this question, we need to study the convergence of Taylor series. To do this, let us consider the n**th-degree Taylor polynomial of** f **at** a:

$$\begin{aligned} T_n(x) &= \sum_{i=0}^{n} \frac{f^{(i)}(a)}{i!}(x-a)^i \\ &= f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f^{(2)}(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n. \end{aligned} \tag{6.7}$$

$\Rightarrow$ first $(n+1)$ terms in Taylor series

Note that the n-th-degree Taylor polynomial is just the sum of the first $(n+1)$ terms in the corresponding Taylor series.

Therefore, $f(x)$ can be represented by its Taylor series if

$$f(x) = \lim_{n\to\infty} T_n(x).$$

We denote the n**-th remainder** of f at a by

$$R_n(x) = f(x) - T_n(x).$$

difference between function and partial sums $\Rightarrow$ the "error"

It follows that

$$f(x) = T_n(x) + R_n(x).$$

So if we can show that the remainder vanishes, i.e. $\lim_{n\to\infty} R_n(x) = 0$, then

$$\lim_{n\to\infty} T_n(x) = f(x).$$

We have therefore proved the following:

Theorem 4. If

$$\lim_{n\to\infty} R_n(x) = 0, \quad \text{for all} \quad |x-a| < R,$$

then f is equal to its Taylor series on the interval $|x-a| < R$.

Our next goal is to study $R_n(x)$.

Theorem 5. (Taylor's Theorem)

Assume that $f^{(n+1)}(x)$ exists and is continuous. Then

hard to compute

$$R_n(x) = \frac{1}{n!}\int_a^x (x-u)^n f^{(n+1)}(u)\, du.$$

it's true for all $n \geq 0$

$$= \int_a^x \frac{f^{(n+1)}(u)\,(x-u)^n}{n!}\, du$$ $\Rightarrow$ very similar to Taylor series

Proof. Set

$$I_n = \frac{1}{n!}\int_a^x (x-u)^n f^{(n+1)}(u)\,du.$$

Approach:

base case

· For $n=0$, we prove it directly from definitions

· For $n>0$, use Integration by-parts

Our goal is to show that $R_n(x) = I_n(x)$.

For $n = 0$, we have by definition that

first term in Taylor series

$R_0(x) = f(x) - T_0(x)$

$$R_0(x) = f(x) - f(a).$$

On the other hand,

f is anti-derivative of f'

$$I_0(x) = \frac{1}{0!}\int_a^x (x-u)^0 f^{(0+1)}(u)\,du = \int_a^x f'(u)\,du = f(x) - f(a),$$ → fundamental theorem of calc

by the Fundamental Theorem of Calculus. Hence, we have proved that

$$R_0(x) = I_0(x).$$

To prove the formula for $n > 0$, we apply Integration by Parts to $I_n(x)$ with

$$U = \frac{(x-u)^n}{n!}, \quad V' = f^{(n+1)}(u).$$

Then

$$U' = -\frac{(x-u)^{n-1}}{(n-1)!}, \quad V = f^n(u).$$

$$\begin{aligned}
I_n(x) &= \int_a^x UV'\,du \\
&= [UV]_a^x - \int_a^x U'V\,du \\
&= \left[\frac{1}{n!}(x-u)^n f^{(n)}(u)\right]_a^x - \int_a^x -\frac{(x-u)^{n-1}}{(n-1)!} f^{(n)}(u)\,du \\
&= -\frac{1}{n!}(x-a)^n f^{(n)}(a) + I_{n-1}(x).
\end{aligned}$$

The result can be rewritten as

$$I_{n-1}(x) = \frac{f^{(n)}(a)}{n!}(x-a)^n + I_n(x) \tag{6.8}$$

Now apply the above recurrence relation n times:

$$
\begin{aligned}
f(x) &= f(a) + I_0(x) \quad \text{(Base case)} \\
&= f(a) + \frac{f'(a)}{1!}(x-a) + I_1(x) \quad \text{by (6.8)} \\
&= f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f^{(2)}(a)}{2!}(x-a)^2 + I_2(x) \quad \text{by (6.8)} \\
&= \vdots \\
&= f(a) + \frac{f'(a)}{1!}(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + I_n(x) \\
&= T_n(x) + I_n(x).
\end{aligned}
$$

This shows that $I_n(x) = f(x) - T_n(x)$ which, by definition, must be equal to $R_n(x)$. □

$R_n(x) = I_n(x)$

Although Taylor's Theorem gives us an explicit formula for the remainder, in practice we normally will not use this formula directly. Instead, we will use it to estimate the size of the error.

local bound

Theorem 6. (Error Bound for Taylor Series)

Let M be a number such that $|f^{(n+1)}(x)| \le M$ for all $|x-a| \le d$. Then

$$|R_n(x)| \le M\frac{|x-a|^{n+1}}{(n+1)!} \qquad \text{for } \; |x-a| \le d.$$

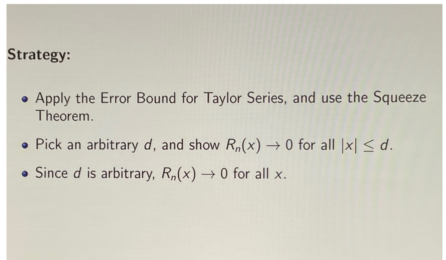
※ M does not depend on n & x.

※ M may depend on d.

Proof. Assume that $a \le x \le a+d$ (the case $a-d \le x \le a$ is similar). Then, since $|f^{(n+1)}(u)| \le M$ for all $a \le u \le x$,

$$
\begin{aligned}
|R_n(x)| &= \left| \frac{1}{n!}\int_a^x (x-u)^n f^{(n+1)}(u)\, du \right| \\
&\le \frac{1}{n!}\int_a^x \left|(x-u)^n f^{(n+1)}(u)\right| du \quad \text{(see Tutorial 1 Question 7)} \\
&\le \frac{M}{n!}\int_a^x (x-u)^n\, du \quad \text{(we ignore the absolute value sign because we assume } a \le u \le x) \\
&= \frac{M}{n!}\left[\frac{-(x-u)^{n+1}}{n+1}\right]_a^x \\
&= M\frac{|x-a|^{n+1}}{(n+1)!}.
\end{aligned}
$$

$\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx$



□

Example 6.12. Prove that $f(x) = e^x$ is equal to its Maclaurin series.

doesn't depend on n

Solution. It is enough to show that the remainder $R_n(x)$ converges to 0 for all x.

Pick an arbitrary positive number d. Note that $|f^{(n+1)}(x)| = e^x \le e^d$ for all $|x| \le d$.

Now apply the Error Bound for Taylor Series with $a = 0$ and $M = e^d$, we have

$$0 \le |R_n(x)| \le \underbrace{\frac{\overset{\to M}{e^d}}{(n+1)!}|x|^{n+1}}_{\to 0} \quad \text{for} \quad |x| \le d.$$

Taking the limit of the right-hand side,

$$\begin{aligned} \lim_{n\to\infty} \frac{e^d}{(n+1)!}|x|^{n+1} &= e^d \lim_{n\to\infty} \frac{|x|^{n+1}}{(n+1)!} \\ &= 0, \end{aligned}$$

where we have used the following fact

$$\lim_{n\to\infty} \frac{x^n}{n!} = 0 \quad \text{for every real number } x. \tag{6.9}$$

→ proved in one of the tutorials

It follows from Squeeze Theorem that $\lim_{n\to\infty} |R_n(x)| = 0$, and so

$$\lim_{n\to\infty} R_n(x) = 0, \quad \text{for} \quad |x| \le d.$$

$-|R_n(x)| \le R_n(x) \le |R_n(x)|$

Since d is arbitrary, the convergence is valid for all x.

Hence, by Theorem 4 and Example 6.11, we have

$$\boxed{e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \text{for all} \quad x.}$$

□

Example 6.13. Find the Maclaurin series for $\sin x$ and prove that it represents $\sin x$ for all $x \in \mathbb{R}$.

Solution. We arrange our computations in the following table:

$$\begin{array}{r|c} f(x) = \sin x & f(0) = 0 \\ f'(x) = \cos x & f'(0) = 1 \\ f^{(2)}(x) = -\sin x & f^{(2)}(0) = 0 \\ f^{(3)}(x) = -\cos x & f^{(3)}(0) = -1 \\ f^{(4)}(x) = \sin x & f^{(4)}(0) = 0 \end{array}$$

Since the derivatives repeat in a cycle of four, we can write the Maclaurin series as follows:

$$\begin{aligned} f(0) + \frac{f'(0)}{1!}x + \frac{f^{(2)}(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \cdots \\ = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \\ = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}. \end{aligned}$$

Next, we will show that $R_n(x) \to 0$.

Since $f^{(n+1)}(x) = \pm \sin x$ or $\pm \cos x$, we have

$$|f^{(n+1)}(x)| \leq 1, \quad \text{for all } x.$$

Apply the Error Bound with $M = 1$, we deduce that

$$|R_n(x)| \leq \frac{M}{(n+1)!}|x^{n+1}| = \frac{|x|^{n+1}}{(n+1)!}.$$

By Equation (6.9), $\lim_{n\to\infty} \frac{|x|^{n+1}}{(n+1)!} = 0$, and so by the Squeeze Theorem, $\lim_{n\to\infty} |R_n(x)| = 0$. It follows that $\lim_{n\to\infty} R_n(x) = 0$ for all x, as required.

By Theorem 4, $\sin x$ is equal to its Maclaurin series for all x:

$$\boxed{\begin{aligned} \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}. \end{aligned}}$$

□

Example 6.14. Find the Maclaurin series for $\cos x$ and prove that it represents $\cos x$ for all $x \in \mathbb{R}$.

Solution. We could proceed directly as in the preceding example, but it's easier to differentiate the Maclarin series for $\sin x$:

$$\begin{aligned}
\cos x &= \frac{d}{dx}(\sin x) \\
&= \frac{d}{dx}\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} = \cdots\right) \\
&= 1 - \frac{3x^2}{3!} + \frac{5x^4}{5!} - \frac{7x^6}{7!} + \cdots \\
&= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots.
\end{aligned}$$

Since the Maclaurin series for $\sin x$ converges for all x, Theorem 2 tells us that the differentiated series for $\cos x$ also converges for all x. Thus

$$\boxed{\begin{aligned}
\cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\
&= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \quad \text{for all} \quad x.
\end{aligned}}$$

Example 6.15. Find the Maclaurin series for the function $f(x) = x \cos x$.

Solution. Instead of computing derivatives from scratch, it's easier to multiply the series for $\cos x$ by x:

maclaurin series for these two functions are valid for all values of x $\Rightarrow$ will definitely converge

converges for x

$$\begin{aligned}
x \cos x &= x \Bigg(\overbrace{\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}}^{\text{converges for } x} \Bigg) \\
&= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n)!}.
\end{aligned}$$

Complicated to calculate $f^{(n)}(x)$.

□

Find the first 4 non-zero terms in the Maclaurin Series for $f(x) = \cos(e^x - 1)$.

$$\begin{aligned}
\cos(e^x - 1) &= \cos\left(\left(1 + x + \frac{x^2}{2!} + \ldots\right) - 1\right) \\
&= \cos\left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\right) \\
&= 1 - \frac{\left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\right)^2}{2!} + \frac{\left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\right)^4}{4!} - \ldots \\
&= 1 - \frac{x^2}{2!} - \frac{1}{2!}\left(x\left(\frac{x^2}{2!}\right) + \left(\frac{x^2}{2!}\right)x\right) - \frac{1}{2!}\left(x \cdot \frac{x^3}{3!} + \frac{x^3}{3!} \cdot x + \left(\frac{x^2}{2!}\right)\left(\frac{x^2}{2!}\right)\right) + \frac{x^4}{4!} + \ldots \\
&= 1 - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4}
\end{aligned}$$

Example 6.16. Write down the first 6 terms of the Taylor series for $\sin x$ at $x = \pi/3$, and prove that the Taylor series represents $\sin x$ for all $x \in \mathbb{R}$.

Solution. Let $f(x) = \sin x$. We arrange our computations in the following table:

$f(x) = \sin x$	$f(\pi/3) = \frac{\sqrt{3}}{2}$
$f'(x) = \cos x$	$f'(\pi/3) = \frac{1}{2}$
$f^{(2)}(x) = -\sin x$	$f^{(2)}(\pi/3) = -\frac{\sqrt{3}}{2}$
$f^{(3)}(x) = -\cos x$	$f^{(3)}(\pi/3) = -\frac{1}{2}$
$f^{(4)}(x) = \sin x$	$f^{(4)}(\pi/3) = \frac{\sqrt{3}}{2}$

Since the derivatives repeat in a cycle of four, we can write the Taylor series as follows:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}\left(\frac{\pi}{3}\right)\left(x-\frac{\pi}{3}\right)^n}{n!} = \frac{\sqrt{3}}{2} + \frac{1}{2}\left(x - \frac{\pi}{3}\right) + \frac{-\sqrt{3}/2}{2!}\left(x - \frac{\pi}{3}\right)^2 + \frac{-1/2}{3!}\left(x - \frac{\pi}{3}\right)^3$$
$$+ \frac{\sqrt{3}/2}{4!}\left(x - \frac{\pi}{3}\right)^4 + \frac{1/2}{5!}\left(x - \frac{\pi}{3}\right)^5 + \cdots$$

Next, we will show that the remainder $R_n(x) \to 0$.

Since $f^{(n+1)}(x) = \pm \sin x$ or $\pm \cos x$, we know that

$$|f^{(n+1)}(x)| \leq 1, \quad \text{for all } x.$$

Apply the Error Bound with $M = 1$, we deduce that

$$|R_n(x)| \leq \frac{M}{(n+1)!}\left|\left(x - \frac{\pi}{3}\right)^{n+1}\right| = \frac{\left|\left(x - \frac{\pi}{3}\right)^{n+1}\right|}{(n+1)!}.$$

constant

$$\because \frac{c^n}{n!} \to 0$$

By Equation (6.9), $\lim_{n\to\infty} \frac{\left|\left(x-\frac{\pi}{3}\right)^{n+1}\right|}{(n+1)!} = 0$, and so by the Squeeze Theorem, $\lim_{n\to\infty} |R_n(x)| = 0$. It follows that $\lim_{n\to\infty} R_n(x) = 0$ for all x, as required.

By Theorem 3, $\sin x$ is equal to its Taylor series at $\pi/3$ for all $x \in \mathbb{R}$. □

Taylor series converges to $\sin x$ for all x.

Let P be an approximation of π accurate to n decimal places.
Show that $p + \sin(p)$ is an approximation to π correct to $3n$ decimal places.

Let's use the Taylor Series for $\sin x$ centred at π.

$$\sin x = -1(x-\pi)^1 + \frac{1(x-\pi)^3}{3!} - \frac{(x-\pi)^5}{5!} + \frac{(x-\pi)^7}{7!} - \cdots$$

Given: $|p-\pi| < \frac{1}{10^n}$

WTS:

$$|p+\sin(p)-\pi| < \frac{1}{10^{3n}}$$

small error

Write $p = \pi + \varepsilon$.

$$|p+\sin p - \pi| = |\pi+\varepsilon+\sin(\pi+\varepsilon)-\pi|$$
$$= \left|\varepsilon+(-\varepsilon)+\frac{\varepsilon^3}{3!}-\frac{\varepsilon^5}{5!}+\ldots\right|$$
$$= \left|\frac{\varepsilon^3}{3!}-\frac{\varepsilon^5}{5!}+\frac{\varepsilon^7}{7!}+\ldots\right| < \frac{|\varepsilon|^3}{3!} < \frac{\left(\frac{1}{10^n}\right)^3}{3!} < \frac{1}{10^{3n}}$$

6.4 The Binomial Series

Reference: Stewart's Calculus 8E, Section 11.10

Goal: To find the Maclaurin series for the function $f(x) = (1+x)^k$ where k is a real number.

Previously: $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad |x|<1 \Rightarrow (1+x)^{-1} = \frac{1}{1+x} = \sum_{n=0}^{\infty} (-x)^n$

$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1} \quad |x|<1 \Rightarrow (1+x)^{-2} = \frac{1}{(1+x)^2} = \sum_{n=1}^{\infty} n(-x)^{n-1}$

Consider expanding the expression $(a+b)^n$, where n is some positive integer. The famous **Binomial Theorem** states that for any positive integer n,

$$\boxed{(a+b)^n = \sum_{i=0}^{n} \binom{n}{i} a^{n-i} b^i,}$$

where $\binom{n}{i}$ is the **binomial coefficient** defined by $0 \le i \le n$

$$\binom{n}{i} = \frac{n!}{i!(n-i)!} = \frac{n(n-1)(n-2)\cdots(n-i+1)}{i!}.$$

For example, for $n = 3$, we have

$$(a+b)^3 = \sum_{i=0}^{3} \binom{3}{i} a^{3-i} b^i = a^3 + 3a^2b + 3ab^2 + b^3.$$

pascal's coef.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

What happens if n is not a positive integer? Can we still apply the Binomial Theorem? To answer this question, we first look at the Maclaurin series for $(1+x)^k$, where k is any real number. For notational convenience, we extend the definition of the binomial coefficient to any real number k:

$$\binom{k}{n} = \frac{\overbrace{k(k-1)(k-2)\cdots(k-n+1)}^{n\text{ terms}}}{n!}, \quad n \ge 1.$$

By convention, $\binom{k}{0} = 1$. Note that if k is a positive integer, this reduces to the usual binomial coefficient.

Example 6.17. Find the Maclaurin series for $f(x) = (1+x)^k$, where k is any real number.

centre is 0

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

<u>*Solution.*</u> We arrange our computations in the following table:

$$\begin{array}{c|c} f(x) = (1+x)^k & f(0) = 1 \\ f'(x) = k(1+x)^{k-1} & f'(0) = k \\ f^{(2)}(x) = k(k-1)(1+x)^{k-2} & f^{(2)}(0) = k(k-1) \\ f^{(3)}(x) = k(k-1)(k-2)(1+x)^{k-3} & f^{(3)}(0) = k(k-1)(k-2) \\ \vdots & \\ f^{(n)}(x) = k(k-1)\cdots(k-n+1)(1+x)^{k-n} & f^{(n)}(0) = k(k-1)\cdots(k-n+1). \end{array}$$

Therefore, the Maclaurin series of $f(x) = (1+x)^k$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = \sum_{n=0}^{\infty} \frac{k(k-1)\cdots(k-n+1)}{n!}x^n = \sum_{n=0}^{\infty} \binom{k}{n} x^n.$$

k can be any real number
doesn't have to be
positive integer

This is called the **Binomial series**. □

$\binom{k}{n} \Rightarrow$ if $n>k$, binomial equation become zero

NOTE: If k is a nonnegative integer, then the terms in the binomial series are eventually 0 and so the series is finite and thus is convergent. For other values of k none of the term is 0 and so we can try the Ratio Test to determine its convergence.

Let $a_n = \binom{k}{n} x^n$. Then

$\binom{k}{k+1} = \frac{k(k-1)\cdots\overbrace{(k-(k+1)+1)}^{0}}{(k+1)!} = 0$

$$\begin{aligned} \left|\frac{a_{n+1}}{a_n}\right| &= \left|\frac{k(k-1)\cdots(k-n+1)(k-(n+1)+1)x^{n+1}}{(n+1)!} \cdot \frac{n!}{k(k-1)\cdots(k-n+1)x^n}\right| \\ &= \frac{|k-n|}{n+1}|x| \\ &= \frac{\left|1-\frac{k}{n}\right|}{1+\frac{1}{n}}|x| \\ &\longrightarrow |x| \quad \text{as} \quad n \longrightarrow \infty. \end{aligned}$$

ρ in Ratio test

Thus, by Ratio Test, the binomial series

R=1

- converges if $|x| < 1$
- diverges if $|x| > 1$.

$\Rightarrow$ inconclusive if $\rho = 1$

The following theorem states that the function $(1+x)^k$ is indeed equal to its Maclaurin series.

Theorem 7. (The Binomial Series)

If k is any real number and $|x| < 1$, then

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n.$$

$k = \frac{1}{2}$ Binomial series converges to $(1+x)^{\frac{1}{2}}$ at $x=1$ $\sqrt{2}$

$k=-2$ Binomial series diverges at $x=1$ [i.e. does not converge to $(1+x)^{-2}$]

NOTE: Although the binomial series always converges when $|x| < 1$, the question of whether or not it converges at the endpoints ± 1, depends on the value of k.

Proof of binomial not tested.

Example 6.18. Find the Maclaurin series for the function $f(x) = \dfrac{1}{\sqrt{4-x}}$ and its radius of convergence.

Solution. We rewrite $f(x)$ in a form where we can apply the binomial series:

$$\begin{aligned} \frac{1}{\sqrt{4-x}} &= \frac{1}{\sqrt{4\left(1-\frac{x}{4}\right)}} \\ &= \frac{1}{2\sqrt{1-\frac{x}{4}}} \\ &= \frac{1}{2}\left(1-\frac{x}{4}\right)^{-1/2}. \end{aligned}$$

Use the binomial series with $k = -\frac{1}{2}$, and with x replaced by $-\frac{x}{4}$, we have

radius of convergence $R=4$

valid provided $|-\frac{x}{4}|<1 \Rightarrow$ i.e $|x|<4$

$$\begin{aligned} \frac{1}{\sqrt{4-x}} &= \frac{1}{2}\sum_{n=0}^{\infty}\binom{-\frac{1}{2}}{n}\left(-\frac{x}{4}\right)^n = \frac{1}{2}\sum_{n=0}^{\infty}\binom{-\frac{1}{2}}{n}(-1)^n\frac{x^n}{4^n} && (6.10) \\ &= \frac{1}{2} + \frac{1}{2}\sum_{n=1}^{\infty}\binom{-\frac{1}{2}}{n}(-1)^n\frac{x^n}{4^n} && (6.11) \end{aligned}$$

by right expansion cannot apply for $n=0$.

We expand $\binom{-\frac{1}{2}}{n}$, for $n \geq 1$, as follows:

$$\begin{aligned} \binom{-\frac{1}{2}}{n} &= \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{2n-1}{2}\right)}{n!} \\ &= (-1)^n\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2^n n!} \end{aligned}$$

Substituting this back into Equation (6.11), we get

$$\frac{1}{\sqrt{4-x}} = \frac{1}{2} + \frac{1}{2}\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n! 8^n} x^n.$$

From the Binomial Series Theorem, we know that this series converges when

$$\left| -\frac{x}{4} \right| < 1,$$

that is $|x| < 4$, so the radius of convergence is $R = 4$. $\square$

6.5 Finding limits using power series

Reference: Stewart's Calculus 8E, Section 11.10

Goal: Use power series to evaluate the limit of a function.

Some important Maclaurin series:

Provided in Exam

$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$	$R = 1$
$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	$R = \infty$
$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$	$R = \infty$
$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$	$R = \infty$
$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$	$R = 1$
$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$	$R = 1$
$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \cdots$	$R = 1$

Example 6.19. Find the sum of the series

$$\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 2^2} + \frac{1}{3 \cdot 2^3} - \frac{1}{4 \cdot 2^4} + \cdots$$

$$= 1 \cdot \tfrac{1}{2} - \tfrac{1}{2}\left(\tfrac{1}{2}\right)^2 + \tfrac{1}{3}\left(\tfrac{1}{2}\right)^3 - \tfrac{1}{4}\left(\tfrac{1}{2}\right)^4 + \ldots$$

$$= 1 . x - \tfrac{1}{2}x^2 + \tfrac{1}{3}x^3 - \tfrac{1}{4}x^4 \qquad x = \tfrac{1}{2}$$

$$= \ln(1+x) = \ln\left(1+\tfrac{1}{2}\right) = \ln\left(\tfrac{3}{2}\right)$$

<u>*Solution.*</u> Using sigma notation, we can write the series as

$$\sum_{n=1}^{\infty}(-1)^{n-1}\frac{1}{n\cdot 2^n}=\sum_{n=1}^{\infty}(-1)^{n-1}\frac{(1/2)^n}{n}.$$

From the table above, this series matches the entry for $\ln(1+x)$ with $x=\frac{1}{2}$. So

$$\sum_{n=0}^{\infty}(-1)^{n-1}\frac{1}{n\cdot 2^n}=\ln(1+\frac{1}{2})=\ln\frac{3}{2}.$$

□

Example 6.20. Use power series to evaluate $\lim_{x\to 0}\frac{e^x-1-x}{x^2}$.

technically can use L'Hopital

$$= \lim_{x\to 0}\frac{e^x-1}{2x}$$

$$= \lim_{x\to 0}\frac{e^x}{2}$$

$$= \frac{e^0}{2}=\frac{1}{2}$$

<u>*Solution.*</u>

$$\begin{aligned}
\lim_{x\to 0}\frac{e^x-1-x}{x^2} &= \lim_{x\to 0}\frac{\left(1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots\right)-1-x}{x^2}\\
&= \lim_{x\to 0}\frac{\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}\cdots}{x^2}\\
&= \lim_{x\to 0}\left(\frac{1}{2!}+\frac{x}{3!}+\frac{x^2}{4!}+\cdots\right)\\
&= \frac{1}{2}.
\end{aligned}$$

□

Example 6.21. Use power series to evaluate $\lim_{x\to 0}\frac{1-\cos x}{1+x-e^x}$. → also can L'Hopital

$$= \lim_{x\to 0}\frac{\sin x}{1-e^x}$$

$$= \lim_{x\to 0}\frac{\cos x}{-e^x}$$

$$= \frac{1}{-1}$$

$$= -1$$

<u>*Solution.*</u> *Give it a try!*

$$\begin{aligned}
\lim_{x\to 0}\frac{1-\cos x}{1+x-e^x} &= \lim_{x\to 0}\frac{1-\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}-\ldots\right)}{1+x-\left(1+x+\frac{x^2}{2!}+\ldots\right)}\\
&= \lim_{x\to 0}\frac{\frac{x^2}{2!}-\frac{x^4}{4!}+\ldots}{-\frac{x^2}{2!}-\frac{x^3}{3!}-\ldots}\\
&= \lim_{x\to 0}\frac{x^2\left(\frac{1}{2!}-\frac{x^2}{4!}+\ldots\right)}{x^2\left(-\frac{1}{2!}-\frac{x}{3!}-\ldots\right)}\\
&= \frac{\frac{1}{2!}-0+0-\ldots}{-\frac{1}{2!}-0-0\ldots}\\
&= -1
\end{aligned}$$